Responsible Machine Learning
Assignment 4
10 points

November 3, 2026

In Assignment 4, you will work with your group to red team your best model following the instructions below and treating your best model as a black box. A template has been provided with examples of simple model extraction and adversarial example attacks.

Please let me know immediately if you find typos or mistakes in this assignment or related materials.

1 Conduct a white-hat model extraction attack.

Cells 10–16 demonstrate a simple, but effective, model extraction attack. My example model extraction attack uses a single decision tree, that I then plot and use to craft adversarial examples. Please conduct a decision tree extraction attack, but you don't have to use my code. (Getting `graphviz` installed could be difficult for some, so feel free to use your favorite kind of decision tree if the template code proves difficult to run.)

You may call `predict()` on your best model only one time to perform the extraction attack.

2 Find adversarial examples for your model.

Cells 17–20 use the information from the extracted decision tree to craft highly effective adversarial examples. You must create at least one adversarial example (row of data) that can evoke low predictions from your best model, and at least one adversarial example that can evoke high predictions.

You may call `predict()` on your best model only one time to test your adversarial examples.

3 Submit Code Results.

Your deliverable for this assignment is to update your group's GitHub repository to reflect this red-teaming exercise. The model extraction attack is worth 5 points, and the adversarial examples are worth another 5 points.

In the real world, after performing this red-teaming exercise, you would want to contact your manager and your organization's IT security team to discuss any discovered vulnerabilities. Countermeasures to discuss with business and IT colleagues may relate to authentication on the vulnerable model API endpoint, throttling/rate-limiting of the vulnerable model API and monitoring the model's production scoring queue for random data and training data, if possible.

Your deliverables are due Sunday, April 13th, at 11:59 PM ET.

Note that you may also improve Assignment 1 or 3 scores to improve your ranking. Moving forward, you'll need to be able to show that your new predictions preserve AIR > 0.8 for all protected groups.